LIMITS OF CONSTRUCTIBLY CLOSED SUBSETS OF SPECTRAL SPACES ARE QUASI-COMPACT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that an inverse limit of quasi-compact spaces is quasi-compact. Invoked inside the counted proof (source lines 5016-5018). Gives quasi-compactness of the constructible limit.
- The statement that the constructible topology of a spectral space is quasi-compact Hausdorff and that spectral maps are continuous for it. Invoked inside the counted proof (source lines 5012-5015). Puts the system in the setting of quasi-compact Hausdorff spaces.
- The statement that quasi-compactness transfers across a surjective continuous map. Invoked at the end of the counted proof (source line 5022). Transfers quasi-compactness from the constructible limit to the limit.

Source reference. Original lemma lemma-closed-in-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1879–1879. Navigation pointer only: the counted text refers to the Stacks item lemma-closed-in-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-constructible-hausdorff-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 4522–4522. Navigation pointer only: the counted text refers to the Stacks item lemma-constructible-hausdorff-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-image-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1964–1964. Navigation pointer

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Source reference. Original lemma lemma-inverse-limit-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 2692–2692. Navigation pointer only: the counted text refers to the Stacks item lemma-inverse-limit-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-limits: sources/stacks-topology/pinned-original-source.tex, source lines 2575–2575. Navigation pointer only: the counted text refers to the Stacks item lemma-limits, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{I}$ be a category. Let $i \mapsto X_i$ be a diagram of spectral spaces such that for $a : j \to i$ in $\mathcal{I}$ the corresponding map $f_a : X_j \to X_i$ is spectral.*

(1) *Given subsets $Z_i \subset X_i$ closed in the constructible topology with $f_a(Z_j) \subset Z_i$ for all $a : j \to i$ in $\mathcal{I}$, then $\lim Z_i$ is quasi-compact.*
(2) *The space $X = \lim X_i$ is quasi-compact.*

Proof. The limit $Z = \lim Z_i$ exists by Lemma lemma-limits. Denote X_i' the space X_i endowed with the constructible topology and Z_i' the corresponding subspace of X_i'. Let $a : j \to i$ in $\mathcal{I}$ be a morphism. As f_a is spectral it defines a continuous map $f_a : X_j' \to X_i'$. Thus $f_a|_{Z_j} : Z_j' \to Z_i'$ is a continuous map of quasi-compact Hausdorff spaces (by Lemmas lemma-constructible-hausdorff-quasi-compact and lemma-closed-in-quasi-compact). Thus $Z' = \lim Z_i$ is quasi-compact by Lemma lemma-inverse-limit-quasi-compact. The maps $Z_i' \to Z_i$ are continuous, hence $Z' \to Z$ is continuous and a bijection on underlying sets. Hence Z is quasi-compact as the image of the surjective continuous map $Z' \to Z$ (Lemma lemma-image-quasi-compact). □